# CHECKING QUASI-COHERENCE OF A COMPLEX ON ETALE COVERS

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: the counted statement is the author lemma environment labelled lemma-check-quasi-coherence-on-covering; the proof environment immediately following it is its own proof; the context span delivers the section definition and the etale pullback discussion that the proof's phrase 'the discussion preceding the lemma' refers to

*Editorial note.* The label lemma-check-quasi-coherence-on-covering names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Allowed standard prerequisites (editorial).*

- the derived category D_QCoh(O_X) and the flat etale pullback facts of the section (delivered as context)
- the characterization of quasi-coherence under etale morphisms (spaces-properties-lemma-characterize-quasi-coherent of Properties of Spaces)

*Source reference.* Original equation 06UR: sources/stacks-chow/tags.txt, source lines 8755–8755. Prerequisite pointer invoked at 592; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 015F: sources/stacks-chow/tags.txt, source lines 1506–1506. Prerequisite pointer invoked at 604; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original section 06UP: sources/stacks-chow/tags.txt, source lines 8753–8753. Prerequisite pointer invoked at 584; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 071Q: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 402–402. Prerequisite pointer invoked at 564; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 04JC: sources/stacks-chow/tags.txt, source lines 5933–5933. Prerequisite pointer invoked at 601; it names a standard fact of a sibling

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chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 073C: sources/stacks-chow/tags.txt, source lines 9056–9056. Prerequisite pointer invoked at 599; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original equation 03LW: sources/stacks-chow/tags.txt, source lines 4728–4728. Prerequisite pointer invoked at 610; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 03M0: sources/stacks-chow/tags.txt, source lines 4732–4732. Prerequisite pointer invoked at 558; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 03M1: sources/stacks-chow/tags.txt, source lines 4733–4733. Prerequisite pointer invoked at 582; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Let $S$ be a scheme. Lemma 071Q shows that the category $D_{QCoh}(\mathcal{O}_S)$ can be defined in terms of complexes of $\mathcal{O}_S$-modules on the scheme $S$ or by complexes of $\mathcal{O}$-modules on the small étale site of $S$. Hence the following definition is compatible with the definition in the case of schemes.

**Definition 0.1.** Let $S$ be a scheme. Let $X$ be an algebraic space over $S$. The *derived category of $\mathcal{O}_X$-modules with quasi-coherent cohomology sheaves* is denoted $D_{QCoh}(\mathcal{O}_X)$.

This makes sense by Properties of Spaces, Lemma 03M1 and Derived Categories, Section 06UP. Thus we obtain a canonical functor

$$D(QCoh(\mathcal{O}_X)) \longrightarrow D_{QCoh}(\mathcal{O}_X) \tag{0.1.1}$$

see Derived Categories, Equation (06UR).

Observe that a flat morphism $f : Y \to X$ of algebraic spaces induces an exact functor $f^* : Mod(\mathcal{O}_X) \to Mod(\mathcal{O}_Y)$, see Morphisms of Spaces, Lemma 073C and Modules on Sites, Lemma 04JC. In particular $Lf^* : D(\mathcal{O}_X) \to D(\mathcal{O}_Y)$ is computed on any representative complex (Derived Categories, Lemma 015F). We will write $Lf^* = f^*$ when $f$ is flat and we have $H^i(f^*E) = f^*H^i(E)$ for $E$ in $D(\mathcal{O}_X)$ in this case. We will use this often when $f$ is étale. Of course in the étale case the pullback functor is just the restriction to $Y_{étale}$, see Properties of Spaces, Equation (03LW).

**Lemma 0.2.** *Let $S$ be a scheme. Let $X$ be an algebraic space over $S$. Let $E$ be an object of $D(\mathcal{O}_X)$. The following are equivalent*

1. *$E$ is in $D_{QCoh}(\mathcal{O}_X)$,*
2. *for every étale morphism $\varphi : U \to X$ where $U$ is an affine scheme $\varphi^*E$ is an object of $D_{QCoh}(\mathcal{O}_U)$,*
3. *for every étale morphism $\varphi : U \to X$ where $U$ is a scheme $\varphi^*E$ is an object of $D_{QCoh}(\mathcal{O}_U)$,*
4. *there exists a surjective étale morphism $\varphi : U \to X$ where $U$ is a scheme such that $\varphi^*E$ is an object of $D_{QCoh}(\mathcal{O}_U)$, and*
5. *there exists a surjective étale morphism of algebraic spaces $f : Y \to X$ such that $Lf^*E$ is an object of $D_{QCoh}(\mathcal{O}_Y)$.*

*Proof.* This follows immediately from the discussion preceding the lemma and Properties of Spaces, Lemma 03M0. □